Flexible Manipulators

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Synonyms

Elastic joint manipulator; Elastic joint robot; Elastic manipulator; Elastic robot; Flexible joint manipulators; Flexible joint robots; Flexible link manipulators; Flexible link robots; Flexible robots

Definition

Flexible Manipulators are robotic systems whose behavior is affected by parasitic elastic effects in a non-negligible extent.

Overview

Classic theory of robotic modeling and control is developed under the assumption that a robot can be described as a simple interconnection of rigid bodies, its behavior being thus dominated by the interaction of actuation and dynamic forces. This hypothesis holds true for a wide range of practically meaningful applications, as proven by the effectiveness of techniques based on it. Nonetheless, this is not always the case, and parasitic structural elasticity can arise in robotic systems for a series of relatively common reasons. As a first example, strain wave gears are a widespread source of structural flexibility – see, e.g., the Harmonic Drive in Fig. 1a. The very working principle of these mechanisms leverages the elasticity of a rotating metal component, called flexspline, which can be modeled as a lumped spring decoupling the motor from the link that it actuates. Flexible transmissions – used for remoting actuators from links – are another commonly used mechanical component introducing elasticity at the joint level. Examples are tendons and elastic belts. These additional moving parts would in principle require their own separate continuum state description, since their mass and elasticity are distributed along the whole mechanism. Yet, their inertia is so low if compared to motor and link ones, that dynamic forces can be neglected, and the distributed elasticity approximated with an equivalent spring. Pneumatic actuation (Fig. 1b) also introduces elasticity acting at joint level, due to the mechanic characteristics of the compressed fluid. Alternatively, a link of the robot can no more be considered as a single rigid body when it is built using lightweight materials or slender designs are employed, and its distributed flexibility must be taken into account (see Fig. 1c). As it will be discussed later, this kind

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(a) Harmonic Drive (b) Pneumatic actuator (c) Lightweigth links

Flexible Manipulators, Fig. 1 Examples of mechanical solutions generating parasitic flexible behavior. Panel (**a**) shows the Harmonic Drive®Strain Wave Gear. (Picture courtesy of Harmonic Drive SE). Panel (**b**) depicts a pneumatic actuator by Festo, Inc. These components introduce elasticity at joint level. Panel (**c**) shows the 2 DOF Serial Flexible Link. (Courtesy of Quanser Inc.), a planar robot with slender design

of flexibility produces fundamentally different dynamic behaviors if compared to joint level one. For the very same reasons, also the robot base can be flexible. Nonetheless, a lumped description of elasticity is typically enough to mathematically describe this condition.

All these sources of elasticity contribute to a rich dynamic behavior that – if neglected – can bring to a strong degradation of transient and overall performance. More specifically, overshooting, undershooting, and small-amplitude/high-frequency oscillations tend to arise, possibly leading to instability of the closed loop system. This in turn can reflect into wrong requirement specifications for the hardware components, However, it would be profoundly wrong to regard these mechanical solutions as examples of bad design. Quite the contrary, they come with substantial advantages making their use extremely attractive. Both the relocation of the actuators closer to the robot base and the more slender design of the robot links go in the direction of reducing the weight of the robot – with particular focus on the terminal parts. This translates in a reduction of the cost of actuation, and in an increased safety in human robot interaction. Furthermore, the overall decreasing of weight is a sensitive goal for those applications where cost of transport is a requirement – as, for example, in space robotics. Finally, strain wave gearings are widely used in robotics for their great advantages if compared to standard ones, in terms of high reduction ratios, compactness, lightweightness, and lack of backlash.

Destabilizing Effect of Flexible Transmissions: A Toy Example

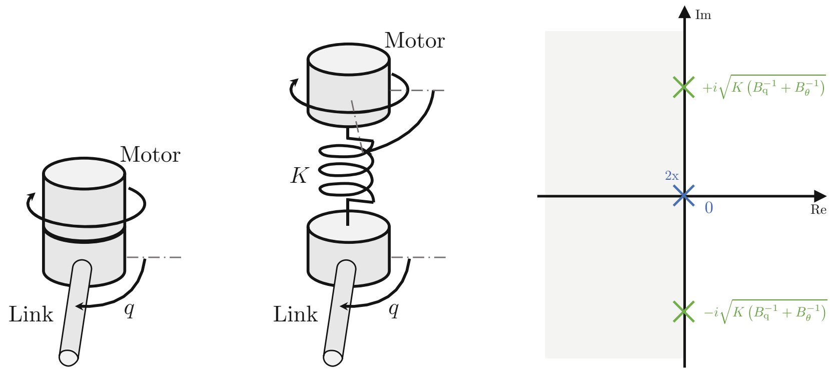
Consider the mechanical system in Fig. 2a, comprising a planar robotic link free to rotate around a fixed axis, and a motor being rigidly connected to the main body. The inertias of the two bodies are B_{q} and B_θ, respectively. A torque τ can be freely applied to the motor. The behavior of this system is described by the ordinary differential equation $(B_{\mathrm{q}} + B_\theta)\ddot{q} = \tau$. In input-output frequency response has two poles in the origin, as shown in Fig. 2c. They are associated to the link rotating with constant speed around its axis. The configuration $q \in \mathbb{R}$ can be regulated through the PD action

$$\tau = -K_{\mathrm{P}} q - K_{\mathrm{D}} \dot{q}, \tag{1}$$

which produces a stable asymptotic behavior for all positive values of the control gains $K_{\mathrm{P}}, K_{\mathrm{D}} \in \mathbb{R}$. This can be tested by direct inspection of the transfer function of the closed loop system $G(s) = 1/\left[(B_{\mathrm{q}} + B_\theta)s^2 + K_{\mathrm{D}} s + K_{\mathrm{P}}\right]$.

Considering the connection between motor and link as not being infinitely rigid yields the simple flexible mechanism shown in Fig. 2b. There the elastic transmission is modeled as a revolute spring decoupling the motor from the link. This system has now four states, and it is described by the linear equations

$$B_{\mathrm{q}}\ddot{q} + K(q-\theta) = 0, \qquad B_\theta\ddot{\theta} + K(\theta - q) = \tau, \tag{2}$$



(a) Rigid (b) Flexible (c) Poles

Flexible Manipulators, Fig. 2 A toy example of flexible mechanism. Panel (**a**) shows a single planar link rigidly connected to the motor actuating it. In panel (**b**), the connection is made flexible and modeled through a linear revolute spring. Two separate variables are required to describe the whole system. Panel (**c**) shows the poles of the transfer function from the input τ (motor torque) to the output q (link configuration). The rigid case has only two poles in the origin. In the flexible case, two extra imaginary poles appear. The first are shown in blue, the second in green

where $\theta \in \mathbb{R}$ is the configuration of the motor, K is the stiffness of the flexible transmission, and the transfer function is $G(s) = 1/\left[(B_\mathrm{q}\,B_\theta/K)\,s^4 + (B_\mathrm{q} + B_\theta)s^2\right]$. A pair of purely imaginary poles appear by effect of the elastic decoupling, as also shown by Fig. 2c. These poles are associated to a purely oscillatory and drift-less mode, modeling the internal vibrations induced by the transmission flexibility. The higher is the stiffness of the mechanism K, the higher is the characteristic frequency of oscillation. The series equivalent of the two inertias appears as a normalizing factor.

It would now seem safe to assume that (1) can stabilize (2), at least for a stiff enough flexible transmission. Yet, this is the transfer function of the closed loop

$$G(s) = 1/\left[(B_\mathrm{q}\,B_\theta/K)\,s^4 + (B_\mathrm{q} + B_\theta)s^2 + K_\mathrm{D}s + K_\mathrm{P}\right]. \qquad (3)$$

Standard tools can be used to prove that (3) *always* describes an unstable system for all the choices of the control gains, no matter how high (but finite) is K. The stability can be recovered by including in τ an action proportional to $\dot{\theta}$. This however requires to separately acquire information on motor and link states with two dedicated sensors. A less sensor demanding alternative is the purely motor side feedback action

$$\tau = -K_\mathrm{P}\theta - K_\mathrm{D}\dot{\theta}. \qquad (4)$$

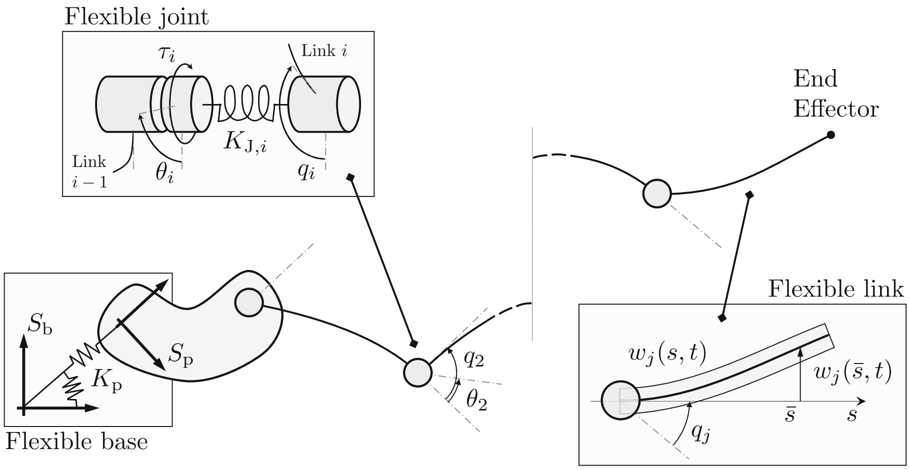
More details on this linear flexible case are provided in Moberg (2010, Ch. 4) and De Luca and Book (2016, Sec 13.1.3).

This example is meant to prove that already the simplest interconnection of masses and flexible elements must be dealt with care. This in turns motivates toward having a closer look at the effects induced by flexible elements in more complex mechanical interconnections.

General Mathematical Structure

Consider a generic serial flexible robot as sketched in Fig. 3 – planar and conservative for the sake of space and readability. The three sources of elasticity discussed above are represented in figure: joint flexibility, link flexibility, and base flexibility.

Already the dynamics of a single flexible link is rather challenging to write down. The model most often taken as a reference in literature is the so-called Euler-Bernoulli elastic beam (see bottom-right window in Fig. 3)



Flexible Manipulators, Fig. 3 Scheme of a generic planar flexible manipulator, with main quantities pointed out. Each link and joint is considered here to be flexible, as well as the robot base. These three sources of elasticity are highlighted in figure in dedicated windows. S_b is a global fixed frame, while S_p is the moving base of the manipulator. K_P is the stiffness matrix connecting the base to the global frame. θ_i is the $i-$th motor angle, and q_i the $i-$th joint angle. The two elements are interconnected through a spring with stiffness $K_{\mathrm{J},i}$. w_j is the shape of the $j-$th link, expressed as the distance from the configuration that would have assumed an equivalent rigid link – parametrized w.r.t. the abscissa s

$$\mu_i(s)\frac{\partial^2 w_i(s,t)}{\partial t^2} + \frac{\partial^2}{\partial s^2}\left(\kappa_i(s)\frac{\partial^2 w_i(s,t)}{\partial s^2}\right) + G_{\mathrm{w},i}(p,q,w,s) = 0,$$
$$H\left(q,\dot{q},\ddot{q},w,\ldots,\frac{\partial^2 w}{\partial t^2}\right) = 0, \tag{5}$$

where $w_i : [0, L_i) \times \mathbb{R} \to \mathbb{R}$ is a smooth function – continuous at least up to the fourth derivative w.r.t. the space coordinate s and up to the second w.r.t. time t – representing the deflection of the $i-$th link from the rest configuration. L_i is the length of the link, and s the coordinate along the link. $q \in \mathbb{R}^{n_\mathrm{J}}$ are the joint angles, and $p \in \mathbb{R}^3$ the posture of the base frame. Being a function of these variables, the gravity field $G_{\mathrm{w},i}$ defines a first coupling between the link dynamics and the rest of the robot. $\mu_i(s) : [0, L_i) \to \mathbb{R}$ is the inertia distribution of the $i-$th link, and $\kappa_i : [0, L_i) \to \mathbb{R}$ is the flexural rigidity – product of the Young modulus and the second moment of inertia of the cross-section.

The boundary constraints $H = 0$ describe how the link is connected to the joints. In the following, it will be assumed that the beginning of each link is rigidly connected to the corresponding joint. In this case the link is said to be clamped. Under this hypothesis, $H = 0$ can be specified as $w(0,t) = 0$, $\frac{\partial w}{\partial t}(0,t) = 0$, $\forall t$. Note that the dynamic couplings between the link deformations and the other parts of the robot state are a by-product of these constraints, and for this reason they do not appear explicitly in (5).

Despite mathematically very sound, building a full model for the flexible robot relying on this link description would make the assessment of structural properties and the design of controllers an arduous task – even if works in literature exist successfully following this path, e.g., by using port-Hamiltonian formalism (Macchelli et al. 2009) or by focusing on simple systems (Chodavarapu and Spong 1996). The model would indeed include multiple infinite dimensional subsystem – one per link – interconnected by means of nonlinear ordinary differential equations (i.e., the dynamics of joints, motors, and base). As an alternative, the largest part of works in the literature considers a simplified description for this system, by transforming the partial differential equations in more tractable ordinary ones. Various discretization techniques can be used to this end, which can be broadly classified in assumed mode (Book 1984) and finite element (Ge et al. 1997) ones. The first rely on re-writing the link configuration as an infinite linear combination of modal oscillations that is then truncated up to a finite order. The second introduce a set of nodes through the beam, which divides the link in virtual elements that can elongate and rotate. The nodes are then connected imposing boundary constraints. An early comparison of the two methods is provided by Theodore and Ghosal (1995). Both techniques lead to a set of ordinary differential equations (ODE), having the same second-order structure of a standard discrete mechanical system.

Merging these ODE with the ones describing joints, motors, and base dynamics yields the following general form

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$$\begin{bmatrix} B_{\theta\theta}(\Theta) & B_{\mathrm{q}\theta}^{\mathrm{T}}(\Theta) & B_{\mathrm{p}\theta}^{\mathrm{T}}(\Theta) & B_{\delta\theta}^{\mathrm{T}}(\Theta) \\ B_{\mathrm{q}\theta}(\Theta) & B_{\mathrm{qq}}(\Theta) & B_{\mathrm{qp}}^{\mathrm{T}}(\Theta) & B_{\delta\mathrm{q}}^{\mathrm{T}}(\Theta) \\ B_{\mathrm{p}\theta}(\Theta) & B_{\mathrm{pq}}(\Theta) & B_{\mathrm{pp}}(\Theta) & B_{\delta\mathrm{p}}^{\mathrm{T}}(\Theta) \\ B_{\delta\theta}(\Theta) & B_{\delta\mathrm{q}}(\Theta) & B_{\delta\mathrm{p}}(\Theta) & B_{\delta\delta}(\Theta) \end{bmatrix} \ddot{\Theta} + \begin{bmatrix} C_{\theta}(\Theta,\dot{\Theta}) \\ C_{\mathrm{q}}(\Theta,\dot{\Theta}) \\ C_{\mathrm{p}}(\Theta,\dot{\Theta}) \\ C_{\delta}(\Theta,\dot{\Theta}) \end{bmatrix} \dot{\Theta}$$

$$+ \begin{bmatrix} K_{\mathrm{J}} & -K_{\mathrm{J}} & 0 & 0 \\ -K_{\mathrm{J}} & K_{\mathrm{J}} & 0 & 0 \\ 0 & 0 & K_{\mathrm{p}} & 0 \\ 0 & 0 & 0 & K_{\delta} \end{bmatrix} \Theta + \begin{bmatrix} G_{\theta}(\Theta) \\ G_{\mathrm{q}}(\Theta) \\ G_{\mathrm{p}}(\Theta) \\ G_{\delta}(\Theta) \end{bmatrix} = \begin{bmatrix} \tau \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{6}$$

where $\delta \in \mathbb{R}^{n_\delta}$ is the finite dimensional counterpart of w, $\theta \in \mathbb{R}^{n_{\mathrm{J}}}$ collects the motor angles, and q and p are defined as above. The four variables are organized in the global configuration of the flexible manipulator $\Theta = \left[\theta^{\mathrm{T}}\ q^{\mathrm{T}}\ p^{\mathrm{T}}\ \delta^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{2n_{\mathrm{J}}+3+n_\delta}$. Dynamic forces are here written in block form to stress the full dynamic coupling within the various components of Θ. $B_{\theta\theta} \in \mathbb{R}^{n_{\mathrm{J}}\times n_{\mathrm{J}}}$, $B_{\mathrm{qq}} \in \mathbb{R}^{n_{\mathrm{J}}\times n_{\mathrm{J}}}$, $B_{\mathrm{pp}} \in \mathbb{R}^{3\times 3}$, $B_{\delta\delta} \in \mathbb{R}^{n_\delta\times n_\delta}$ are the inertia matrices for motors, joints, base dynamics, and link deformations, respectively. $B_{\mathrm{q}\theta} \in \mathbb{R}^{n_{\mathrm{J}}\times n_{\mathrm{J}}}$, $B_{\mathrm{p}\theta} \in \mathbb{R}^{3\times n_{\mathrm{J}}}$, $B_{\mathrm{pq}} \in \mathbb{R}^{3\times n_{\mathrm{J}}}$, $B_{\delta\theta} \in \mathbb{R}^{n_\delta\times n_{\mathrm{J}}}$, $B_{\delta\mathrm{q}} \in \mathbb{R}^{n_\delta\times n_{\mathrm{J}}}$, $B_{\delta\mathrm{p}} \in \mathbb{R}^{n_\delta\times 3}$ account for inertia couplings. $C_\theta \in \mathbb{R}^{n_{\mathrm{J}}\times 2n_{\mathrm{J}}+3+n_\delta}$, $C_{\mathrm{q}} \in \mathbb{R}^{n_{\mathrm{J}}\times 2n_{\mathrm{J}}+3+n_\delta}$, $C_{\mathrm{p}} \in \mathbb{R}^{3\times 2n_{\mathrm{J}}+3+n_\delta}$, $C_\delta \in \mathbb{R}^{n_{\mathrm{J}}\times 2n_\delta+3+n_\delta}$ collect Coriolis and centrifugal effects. $K_{\mathrm{J}} \in \mathbb{R}^{n_{\mathrm{J}}\times n_{\mathrm{J}}}$ is the stiffness matrix – diagonal and positive defined – of the elastic connection between motors and joints. Note that in accordance with the large part of the related literature, only elasticity in the rotational direction is considered here. It should however be mentioned that models exist taking into account other modes of deformation (Moberg et al. 2014). $K_{\mathrm{p}} \in \mathbb{R}^{3\times 3}$ is the semi-positive defined stiffness matrix of the flexible base, which without loss of generality is considered to be at rest in the presence of no external load in $p = 0$. It is worth mentioning here that if $K_{\mathrm{p}} = 0$, then the model corresponds to the floating base case, as, for example, discussed in Englsberger et al. (2014), Yüksel et al. (2016), and Suarez et al. (2018). $K_\delta \in \mathbb{R}^{n_\delta\times n_\delta}$ is the stiffness matrix governing the oscillations of the links. It is positive defined and block diagonal, with each block associated to a different link. All the elastic forces have been considered linear for the sake of simplicity, but most of the results that will be introduced can be generalized to smooth elastic fields. Finally, $G_\theta \in \mathbb{R}^{n_{\mathrm{J}}}$, $G_{\mathrm{q}} \in \mathbb{R}^{n_{\mathrm{J}}}$, $G_{\mathrm{p}} \in \mathbb{R}^{3}$, and $G_\delta \in \mathbb{R}^{n_\delta}$ are the gravity actions. In the following, B, C, $K \in \mathbb{R}^{2n_{\mathrm{J}}+3+n_\delta\times 2n_{\mathrm{J}}+3+n_\delta}$ are used to refer to the full inertia, Coriolis and centrifugal, and stiffness matrices, respectively. Similarly, $G \in \mathbb{R}^{2n_{\mathrm{J}}+3+n_\delta}$ is the full gravity vector.

Control Challenges Distinguishing the Flexible Case from the Rigid One

Four distinct sub-dynamics can be recognized in (6) – namely, $\ddot{\theta}$, $\ddot{q}$, $\ddot{p}$, and $\ddot{\delta}$. The four sets of ODE are interconnected through a complex network of cross-talking effects (summarized by Fig. 4): at the level of accelerations and velocities – mediated by dynamic forces – and through more structured configuration-dependent couplings, mediated by the potential fields. The first are sometimes referred in literature as *fast* couplings, and the second as *slow* couplings.

The only actuated subsystem is the motor dynamics $\ddot{\theta}$. On the other hand, tasks are typically formulated in the other three subsystems, on which the input does not act directly (Chodavarapu and Spong 1996). Examples are the positioning of the end effector, the trajectory tracking of joint space trajectory q, and the damping of oscillations in p and δ. To achieve these goals, it is thus necessary to exploit the coupling effects offered by the dynamics itself. These characteristics can be summarized in the first main challenges making the control of flexible manipulators hard:

C1 Under actuation: less independent inputs are available than degrees of freedom.
C2 Non-collocation: the control objective is specified through variables on which the inputs do not directly act.

Challenges 1 and 2 make a substantially more complex scenario if compared to the classic, fully actuated, rigid robot case – in which the controller has full authority on the dynamics and any behavior can be imposed by direct cancellation and replacement.

A more technical issue, which however is not less important than the first two, is the following:

C3 Non-minimum phase: the zero-dynamics of the non-collocated outputs is often unstable.

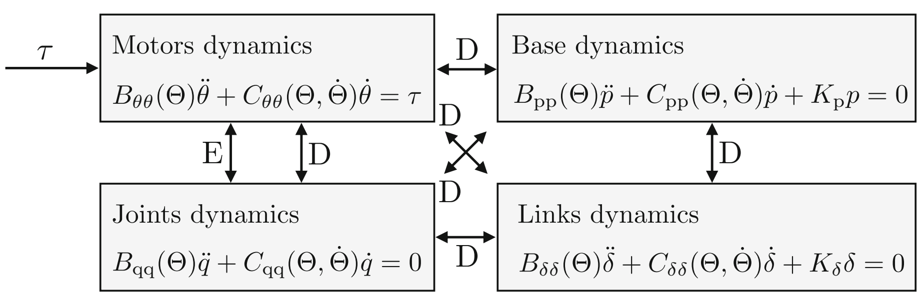
This third challenge has a great impact in the design of effective controllers – as we will discuss later – since it prevents acting on the system through input-output feedback linearization. Finally, the model framed as in (6) is inherently affected by performance and stability issues related to the portion of the link dynamics that is neglected.

C4 Control spillover: the truncated dynamics could be excited by the controller designed on the reduced system.

An early work pointing out this issue is Balas (1978).

Main Properties

Equation (6) is quite comprehensive, and its direct analysis and control is a problem often too complex for being tackled in the general case. One way to attack it is to consider very reduced and simplified – yet meaningful – examples, as a single flexible link with a flexible joint and a fixed base (Wei et al. 2017). Instead, other assumptions are needed to get to more general results. The most commonly studied specific cases will be considered in the following section – obtained by considering separately each of the three sources of elasticity, namely, flexible joint robots, flexible link robots, and flexible base robots.



Flexible Manipulators, Fig. 4 Simplified scheme of interconnection between the four sub-systems building the dynamics of a flexible manipulator. Arrows marked with a D indicate dynamic coupling, while a E refers to elastic coupling. Gravity is not considered in this figure

However, several useful properties can be already recognized, generalizing the corresponding ones from rigid robots.

P1 B is symmetric and positive defined
P2 C can be written such that $\dot{B} - 2C$ is skew-symmetric
P3 $||B||$ can be upper bounded with an affine function of $||\Theta||$.
P4 $||C||$ can be upper bounded with a second-order polynomial in $||\Theta||$ and $||\dot{\Theta}||$.
P5 $||G||$ can be upper bounded with an affine function of $||\Theta||$.

These properties hold true for the sub-cases that are considered in the following. However, adding additional hypotheses on the flexible manipulator brings out further structural characteristics, which are of fundamental importance in the development of appropriate control strategies.

A Note on the Difference Between Soft Robots and Flexible Manipulators

Soft robots are robotic systems having elastic elements purposefully included in their mechanics to implement a desired compliant behavior. Therefore, on a conceptual level, soft and flexible robots differ in having elasticity as a specifically designed behavior in the first and as an unwanted parasitic effect in the seconds. This translates into different requirements on the mechanical components and mechanisms to implement them – as opportunely designed springs and soft materials. In flexible robots, the joint stiffness is in the order of $10^4 \frac{\mathrm{Nm}}{\mathrm{rad}}$, while in articulated soft robots, it is around $10^{-1} \frac{\mathrm{Nm}}{\mathrm{rad}}$. Similarly, continuum soft robots are made of very soft materials, with Young moduli in the range $10^4 \sim 10^8$ Pa, while the range for flexible links is $10^{10} \sim 10^{11}$ Pa.

Despite the different levels of flexibility, on a mathematical level the two classes of robots have strong similarities. This is especially true for articulated robots, which in their series actuated form are described by exactly the same set of ordinary differential equations flexible joint robots are described by (see next section). This circumstance generated some confusion in the soft robotics community, which favored when possible a direct application of techniques developed for flexible manipulators to the articulated soft case. However, the aim of any control architecture for flexible manipulators is to recover the same level of performance that one would achieve with a rigid robot, *despite* the presence of elasticity. This translates into removing the elasticity from the system by closed loop control. Thus, applying controllers devised for flexible manipulators to soft ones results in the stiffening of the whole structure, making useless the introduction of elasticity (Santina et al. 2017). Variable stiffness robots can be considered as an exception, since making them more rigid through control makes sense when this is the behavior implemented by the hardware (Buondonno and Luca 2016).

Key Research Findings

Rather than attacking the problem in the general case introduced in the previous section, the literature often focuses on the three simplified cases arising from the stiffening of two out of the three sources of elasticity. They are discussed in the following, in the order in which they historically appeared in the state of the art.

Flexible Joint Robots

These robots are obtained when joint level elasticity is taken as dominant effect. The dynamics results as limit of (6), for $||K_{\mathrm{p}}||$ and $||K_\delta||$ going to infinity

$$\begin{bmatrix} B_{\theta\theta}(\theta,q) & B_{\mathrm{q}\theta}^{\mathrm{T}}(\theta,q) \\ B_{\mathrm{q}\theta}(\theta,q) & B_{\mathrm{qq}}(\theta,q) \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{q} \end{bmatrix} + \begin{bmatrix} C_\theta(\theta,q,\dot{\theta},\dot{q}) \\ C_{\mathrm{q}}(\theta,q,\dot{\theta},\dot{q}) \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{q} \end{bmatrix} + \begin{bmatrix} K_{\mathrm{J}} & -K_{\mathrm{J}} \\ -K_{\mathrm{J}} & K_{\mathrm{J}} \end{bmatrix} \begin{bmatrix} \theta \\ q \end{bmatrix} + \begin{bmatrix} G_\theta(\theta,q) \\ G_{\mathrm{q}}(\theta,q) \end{bmatrix} = \begin{bmatrix} \tau \\ 0 \end{bmatrix}.$$

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This dynamics is still very general, and the following assumptions are often introduced to further simplifying analysis and control: (i) each motor has its center of mass included in the rotation axis, (ii) each motor is rigidly connected to the link preceding the driven link, with rotation axis aligned with the joint, (iii) the effect of $\dot{q}$ is negligible w.r.t. $\dot{\theta}$ when evaluating the angular speed of the motors. Assumptions (i) and (ii) are true for almost all flexible joint robots, and they already help in removing several couplings between the two subsystems. More specifically C_θ and C_q become a function of q and $\dot{q}$ only, $B_{\theta\theta}$ becomes a constant matrix, $B_{\theta q}$ is function of q only, G_q is function of q only, and G_θ vanishes.

Assumption (iii) is true when motors are equipped with gears with large reduction ratios. While this is a common practice, it is an assumption that does not always hold true. Nonetheless, it is commonly introduced since it produces a strong simplification of the system dynamics, i.e., $B_{\mathrm{q}\theta}(\theta, q) \simeq 0$. Note that this is a quite relevant change, since $B_{\mathrm{q}\theta}(\theta, q)$ was the last non-purely elastic coupling that remained after the introduction of the first two assumptions.

Combining the three assumptions yields the following simplified dynamics

$$
\begin{aligned}
B_{\theta\theta}\ddot{\theta} + K_{\mathrm{J}}(\theta - q) &= \tau, \\
B_{\mathrm{qq}}(q)\ddot{q} + C_{\mathrm{q}}(q, \dot{q})\,\dot{q} + G_{\mathrm{q}}(q) + K_{\mathrm{J}}(q - \theta) &= 0,
\end{aligned} \tag{7}
$$

which are two purely elastically coupled subsystems, one linear and fully actuated and the other not actuated and nonlinear. Note that this is a direct nonlinear extension of the toy model (2).

System (7) can be completely feedback linearized when choosing as control output the joint configuration q. Indeed, differentiating two times this output yields the joint space dynamics. The control input τ appears only after deriving other two times, thanks to the dependence on θ in the elastic coupling, which at the fourth derivation becomes $\ddot{\theta}$. The feedback linearizing controller is

$$
\tau = (B_{\mathrm{qq}} + B_{\theta\theta})\ddot{q} + C_{\mathrm{q}}\dot{q} + B_{\theta\theta}K_{\mathrm{J}}^{-1}\left(B_{\mathrm{qq}}P\begin{bmatrix} q \\ \dot{q} \\ \ddot{q} \\ \dddot{q} \end{bmatrix} + \ddot{B}_{\mathrm{qq}}\ddot{q} + 2\dot{B}_{\mathrm{qq}}\dddot{q} + \frac{\mathrm{d}^2 C_{\mathrm{q}}\dot{q}}{\mathrm{d}t^2} \right),
$$

where we omitted dependencies for the sake of space and readability, and P is a gain matrix to be selected in the usual way so to achieve a desired pole placement of the linearized system. Note that τ is always well defined, for any fully coupled elastic structure, i.e., as soon as K_{J} full rank. It is also worth noticing that, while well defined in theory, it is not necessary to directly measure $\ddot{q}$, $\dddot{q}$. This information can be regressed by more easy to obtain measurements – as θ and $\dot{\theta}$, or $\tau_{\mathrm{e}} = K_{\mathrm{J}}(\theta - q)$ and $\dot{\tau}_{\mathrm{e}}$. The first can be measured with encoders placed on the motor side and the second from torque sensors connected on the link side.

As discussed above, (7) has a cascade structure, which makes viable the use of backstepping (Oh and Lee 1997). This technique directly results into an acceleration-free feedback.

The control problem becomes much harder as soon as hypothesis (iii) is relaxed, i.e., $B_{\mathrm{q}\theta}(\theta, q) \not\simeq 0$. The amount of times that one needs to differentiate q to obtain τ is now just two, since $\ddot{\theta}$ already appears in the dynamics of the joint. This implies that the system can be feedback linearized only partially. Moreover, under assumptions (i) and (ii), the output dynamics is

$$
\begin{aligned}
&B_{\mathrm{qq}}(\theta, q)\ddot{q} - B_{\mathrm{q}\theta}B_{\theta\theta}^{-1}(q)\left(B_{\mathrm{q}\theta}^{\mathrm{T}}(q)\ddot{q} + C_\theta(q, \dot{q})\dot{q} + K_{\mathrm{J}}(\theta - q) - \tau\right) \\
&\quad + C_{\mathrm{q}}(q, \dot{q})\,\dot{q} + G_{\mathrm{q}}(q) + K_{\mathrm{J}}(q - \theta) = 0,
\end{aligned}
$$

where it is easy to prove that $\det\{B_{\mathrm{q}\theta}\} = 0$ is always true. This makes the output dynamics not fully actuated, therefore preventing the direct use of feedback linearization. A way around has been identified in the use of dynamic controllers. Instead of directly producing τ, the control output is filtered through a generic nonlinear systems with dynamics dependent on the full state of the robot. The design of this dynamic system is discussed in De Luca (1988), where it is also proven that the resulting augmented dynamics can be input-output linearized.

A middle ground between the dynamic coupled and the purely elastic coupled cases can be identified in the flexible and damped joint case

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$$B_{\theta\theta}\ddot{\theta} + K_{\mathrm{J}}(\theta - q) + D_{\mathrm{J}}(\dot{\theta} - \dot{q}) = \tau,$$

$$B_{\mathrm{qq}}(\theta, q)\ddot{q} + C_{\mathrm{q}}(q, \dot{q})\,\dot{q} + G_{\mathrm{q}}(q) + K_{\mathrm{J}}(q - \theta) + D_{\mathrm{J}}(\dot{q} - \dot{\theta}) = 0,$$

discussed in De Luca et al. (2005). In this circumstance, the relative degree between input τ and output q is 3, again preventing a complete feedback linearization. However, the static controller can be designed so that the zero dynamics is exponentially stable with any desired decreasing rate. In case of small damping, this static linearizing feedback tends to diverge. This can be corrected resorting to augmenting the dynamics, as for the dynamically coupled case.

Feedback linearization is a valid way of describing the controllability properties of flexible joint robots. Nonetheless, more robust, simpler, less model-based, and less sensor demanding strategies could be preferable in the practice. A quite general solution of the regulation problem is provided in Albu-Schäffer et al. (2007) and can be summarized by the following control law:

$$\tau = \bar{\tau} - K_{\mathrm{T}}(\tau_{\mathrm{e}} - \bar{\tau}) - D_{\mathrm{T}}\dot{\tau}_{\mathrm{e}} - K_{\mathrm{P}}(\theta - \bar{\theta}) - D_{\mathrm{P}}\dot{\theta}, \quad (8)$$

where $\bar{\tau}$ is a desired torque to be applied at joint level, τ_{e} is the same torque measured by on-board sensors, and $K_{\mathrm{T}}, D_{\mathrm{T}}, K_{\mathrm{P}}, D_{\mathrm{P}}$ are gain matrices. When $K_{\mathrm{P}} = 0$ and $D_{\mathrm{P}} = 0$, the feedback action (8) becomes a torque regulator and in this form is used on the KUKA lwr to mask the joint level elasticity. As an alternative, a direct extension of the motor side regulator (4) discussed in section "Destabilizing Effect of Flexible Transmissions: A Toy Example" is obtained by picking $K_{\mathrm{T}} = 0$ and $D_{\mathrm{T}} = 0$. In case the robot is subject to gravity, then $\bar{\tau} = G_{\mathrm{q}}(\bar{q})$ implements gravity compensation (Tomei 1991), with $\bar{q}$ being the desired robot configuration. Also, the motor reference must be adjusted accordingly $\bar{\theta} = \bar{q} + K_{\mathrm{J}}^{-1}G_{\mathrm{q}}(\bar{q})$. If instead all gains are chosen to be not null, then (8) can realize a full-state regulator or an impedance controller. Joint space impedance control is implemented for robots with damped transmissions in Garofalo et al. (2018). Similar control schemes can also be used to implement trajectory tracking (Giusti et al. 2018) rather than simple regulation. This can be achieved by generating feed-forward actions through direct inversion of (7)

$$\bar{\theta} = \bar{q} + K_{\mathrm{J}}^{-1}\left[B_{\mathrm{qq}}(\bar{q})\ddot{\bar{q}} + C_{\mathrm{q}}(\bar{q}, \dot{\bar{q}})\,\dot{\bar{q}} + G_{\mathrm{q}}(\bar{q})\right],$$
$$\bar{\tau} = B_{\theta\theta}\ddot{\bar{\theta}} + K_{\mathrm{J}}(\bar{\theta} - \bar{q}), \quad (9)$$

where the closed form expression for the term $\ddot{\bar{\theta}}$ can always be evaluated analytically. In turn, $\bar{q}$ can be designed so to be characterized by minimum execution time or low jerk (Kim and Croft 2019). A thorough discussion on the use of feedback linearization and feedforward schemes for trajectory tracking in real-world scenarios is provided by Moberg (2010).

As for the rigid case, the flexible joint dynamics admits a linear parametrization in terms of a suitable combination of joint stiffnesses and motor inertias. This property can be exploited for identification purposes (Flacco and De Luca 2011) and in adaptive control. A first result in this direction is Ghorbel et al. (1989), where singular perturbation arguments are used to derive the control law. While possible in theory, evaluating this linear parametrization is not always easy or even feasible in the practice. Regressor-free adaptive schemes (Chien and Huang 2007) and iterative learning control (Ailon 1996) proved to be two valuable alternatives, for increasing tracking performance under uncertainties while not needing any explicit formulation of the regressor. The use of machine learning strategies has been investigated too, as wavelet networks (Yoo et al. 2008) and deep neural networks (Takahashi et al. 2017).

Going beyond configuration space control, Cartesian impedance control is described in Ott (2008), while hybrid force and impedance control are discussed in Schindlbeck and Haddadin (2015).

Flexible Link Robots

These robots are obtained when link flexibility is considered as the dominant effect. The dynamics can be derived as limit of (6), for $||K_\mathrm{p}||$ and $||K_\mathrm{J}||$ going to infinity

$$\begin{bmatrix} B_{\mathrm{qq}}(q,\delta) & B_{\delta\mathrm{q}}^{\mathrm{T}}(q,\delta) \\ B_{\delta\mathrm{q}}(q,\delta) & B_{\delta\delta}(q,\delta) \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \ddot{\delta} \end{bmatrix} + \begin{bmatrix} C_{\mathrm{q}}(q,\delta,\dot{q},\dot{\delta}) \\ C_{\delta}(q,\delta,\dot{q},\dot{\delta}) \end{bmatrix} \begin{bmatrix} \dot{q} \\ \dot{\delta} \end{bmatrix} + \begin{bmatrix} 0 \\ K_\delta \delta \end{bmatrix} + \begin{bmatrix} G_{\mathrm{q}}(q,\delta) \\ G_{\delta}(q,\delta) \end{bmatrix} = \begin{bmatrix} \tau \\ 0 \end{bmatrix}.$$

A characteristic cascade nature can be recognized into this dynamics too. In contrast with the flexible joint case, the coupling between the two subsystems is purely mediated by dynamic forces rather than by potential ones – at least in absence of a gravity field. This, together with the high dimensionality of δ, makes for a much harder control problem. Among the other things, it has been shown that these characteristics imply that directly tracking meaningful outputs related to the end effector position generates non-minimum phase input-output characteristics, leading to unstable closed loop behaviors. See, for example, (Chodavarapu and Spong 1996) and references within.

A general way of regulating joint space variables so that the whole closed loop dynamics is stable is discussed in De Luca and Siciliano (1993), where static feedback linearization is applied using as output q. The zero dynamics $\ddot{\delta}$ is proven to be globally asymptotically stable in the equilibrium. While solving the problem in theory, this control strategy has two main drawbacks.

First, tracking in q space is not practical. Indeed – contrary to the rigid and flexible cases – the task space cannot be completely specified by an equivalent trajectory in joint space, since the link deformations δ contribute to the task outcome. Optimization algorithms can be used to plan an evolution in joint space taking into account and minimizing these effects (Park 2004). As an alternative, tip regulation can be considered instead of tracking. This substantially simpler problem can be tackled by combining joint level feedback schemes (De Luca and Siciliano 1993) with simple kinematic inversion algorithms.

Second, the general strategies that can be derived using feedback linearization are very sensor demanding and strongly model based – so often of impractical implementation. As a result, large part of the research in the field moved to considering less general conditions, so to obtain simpler yet effective control laws. A commonly followed approach is to linearize the link dynamics for small vibrations, possibly integrating it in an hierarchical control architecture – i.e., implementing low-level controllers regulating δ regardless of q. From a theoretical point of view, this approach can be motivated through singular perturbation arguments (Siciliano and Book 1988). In alternative, the whole dynamics can be linearized around a desired configuration or trajectory (De Luca 2000). This approach can be taken even further, using model-free strategies as repetitive/iterative control (De Luca and Panzieri 1994), fuzzy systems, and neural networks (Rahimi and Nazemizadeh 2014).

On the other end of the spectrum, there is a large body of work specifically tailored on the single link case – for which in presence of no gravity and for small link deformations the problem becomes linear. This way of reducing the problem complexity generates simple and effective solutions, which at the same time can retain a deep theoretical interpretation. Examples include all the techniques discussed above for the general case, model predictive control (Dubay et al. 2014), direct control of the infinite dimensional system (Guo 2002), sliding mode (Yeung and Chen 1989), and input shaping (Singhose 2009), just to cite a few.

Finally, an important area of research, transversal to the various techniques discussed above, is modeling. Using the Euler-Bernoulli model requires the introduction of several assumptions. First and foremost the load is considered always orthogonal to the beam – which is never true in the presence of gravity. Moreover, the resulting deformations are hypothesized to be small, and the beam is supposed to undergo pure bending deformations. If these assumptions are not simultaneously fulfilled – as, for example, in the presence of elongation, torsion, or large oscillations – rod theories should be considered instead, as, e.g., proposed in Boyer et al. (2010).

F

Flexible Base Robots

A latter case is the one with rigid robot structure but flexible base. The dynamics can be derived as limit of (6), for $||K_{\mathrm{J}}||$ and $||K_{\delta}||$ going to infinity

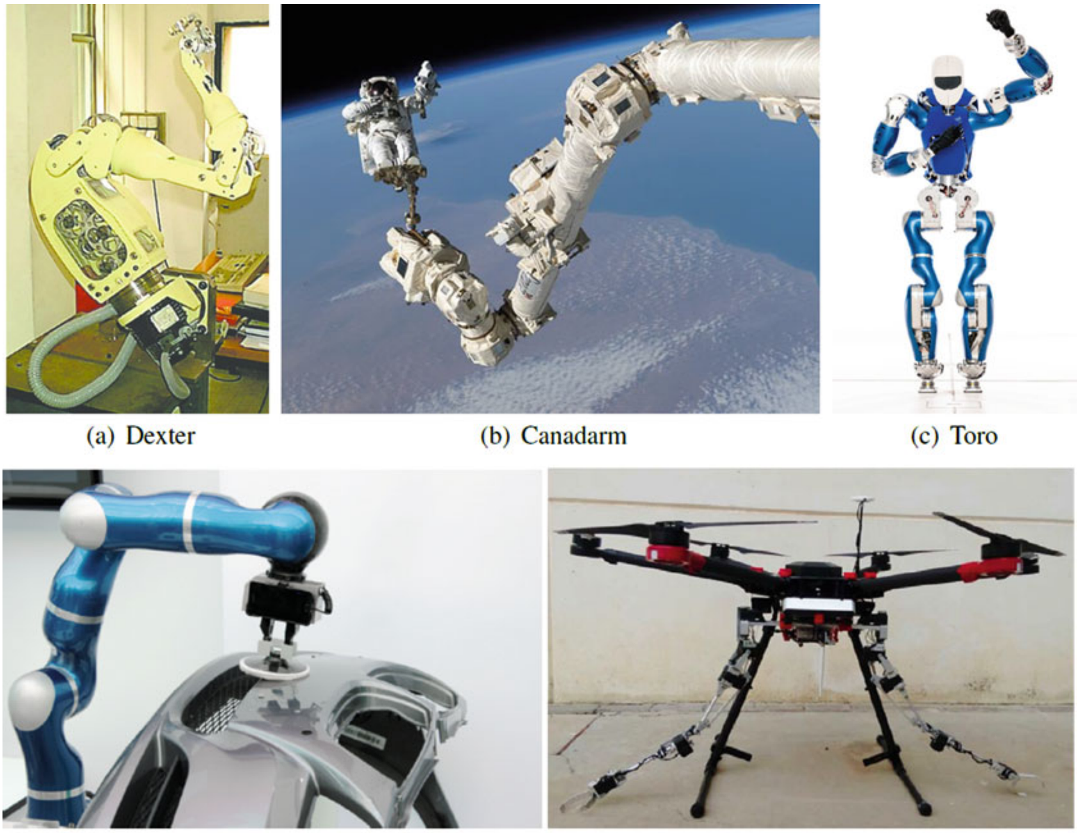
$$\begin{bmatrix} B_{\mathrm{qq}}(q,p) & B_{\mathrm{qp}}^{\mathrm{T}}(q,p) \\ B_{\mathrm{pq}}(q,p) & B_{\mathrm{pp}}(q,p) \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \ddot{p} \end{bmatrix} + \begin{bmatrix} C_{\mathrm{q}}(q,p,\dot{q},\dot{p}) \\ C_{\mathrm{p}}(q,p,\dot{q},\dot{p}) \end{bmatrix} \begin{bmatrix} \dot{q} \\ \dot{p} \end{bmatrix} + \begin{bmatrix} 0 \\ K_{\mathrm{p}}p \end{bmatrix} + \begin{bmatrix} G_{\mathrm{q}}(q,p) \\ G_{\mathrm{p}}(q,p) \end{bmatrix} = \begin{bmatrix} \tau \\ 0 \end{bmatrix}.$$

Note that the subsystems interaction structure is very similar to the one of flexible link robots. However, the two differ for the relative size of the systems. Indeed in the presence of link flexibility $n_{\delta} >> n_{\mathrm{J}}$, while n_{J} is rarely smaller than the three degrees of freedom of the base. This provides to a controller a considerably larger authority on the passive dynamics.

Existing control techniques tackle the goal of achieving end effector regulation, while damping base oscillations. This goal can be reached within a singular perturbation framework, approximating the robot as a combination of a slow rigid dynamics and a fast flexible base dynamics (Book and Lee 1989). Alternatively, the control goal can be achieved exactly and without any need of adding additional hypotheses by exploiting null-space motions of redundant manipulators (Nenchev et al. 1999). More recently, Beck et al. (2019) proved that task accomplishment and base stabilization can be achieved simultaneously with nonredundant robots, in the hypothesis of negligible rotations of the base.

Examples of Application

Technologies producing flexible joints are nowadays widespread in the robotic practice. Classic examples are the PUMA manipulator and Dexter by Scienzia Machinale (Fig. 5a). The first is actuated through long shafts, the second through tendons. The control of the first is discussed in Pfeffer et al. (1989), and the control of the second in Zollo et al. (2003). The control of the IRB4600 manipulator from ABB is instead discussed in Moberg et al. (2014). Another very relevant example of flexible joint robot is the DLR lightweight robot, later evolved in the well-known Kuka lwr (Albu-Schäffer et al. 2007). This manipulator is controlled through (8) to mask its elastic nature, presenting to the user a torque interface. Under this configuration, it finds use in the most wide range of applications, e.g., human robot interaction (Peternel et al. 2018), industrial applications (Schindlbeck and Haddadin 2015), humanoid robotics (Englsberger et al. 2014), and space robotics (Giordano et al. 2019). Some of these applications are summarized by Fig. 5.



(a) Dexter (b) Canadarm (c) Toro (d) Kuka lwr (e) Slender arm

Flexible Manipulators, Fig. 5 Some examples of flexible manipulators used in real-world situations. Panels (**a**,**c**,**d**) show flexible link robots, while panels (**b**,**e**) depict flexible link ones. Panel (**a**) reports Dexter by Scienzia Machinale (Zollo et al. 2003), where motors are remonetized near to the base, thanks to tendon transmissions. Panel (**b**) shows the Canadarm. (Photo courtesy by NASA). Panel (**c**) depicts the flexible joint humanoid Toro (Englsberger et al. 2014). (Photo courtesy by DLR). Panel (**d**) displays a Kuka lwr polishing the front apron of a car (Schindlbeck and Haddadin 2015). Panel (**e**) shows an aerial manipulator equipped with two slender arms (Suarez et al. 2018)

The development and control of flexible link structures is instead still confined to lower technology readiness levels. This is mostly due to the balance between mechanical and increasing in control complexity, which for flexible link robots tends to be more shifted in direction of the second if compared to the flexible joint case. Nonetheless, several meaningful practical implementations of this technology exist. Space applications are the most compelling ones in this sense, since the size of the mechanical structure must be reduced to the bare minimum to maximize the amount of instrumentation that can be transported during a single delivery (Flores-Abad et al. 2014). However, controllers for these systems do no operate any explicit compensation of flexibility, which is left to the skilled operator to deal with. More recently, flexible link robots are being used as arms of aerial manipulators. Indeed, the payload that these systems can carry is typically quite low, and therefore reducing the weight of the added robotic components is paramount (Suarez et al. 2018). An example is shown in Fig. 5e. The need of taking into account flexible links also arises when one or more robots are carrying a flexible load (Yang et al. 2018).

Aerial manipulators are revitalizing also interest in flexible base systems (Fumagalli et al. 2014). When a floating base system continuously interacts with or hold at the environment, the contact can be modeled as a lumped spring, and it assumes the role of a flexible base. This situation can arise during inspection and maintenance missions, as well as when the robot must resist to external disturbances – e.g., a strong wind. More classic systems producing a flexible base are micro-macro robots, when the joints of the macro part are blocked (George and Book 2002). These systems are used in chemical and nuclear facilities. The macro part of the robot is used to reach restricted areas, where then the micro part operates.

Future Directions for Research

Nowadays, flexible robotics can be regarded as a mature area of research, with several profound theoretical and practical outcomes and more than 30 years of active research. Among flexible robots, flexible joint ones are the more deeply understood from a theoretical standpoint, with widespread real-world applications. On the contrary, flexible link robots remain a more active area of research due to the inherent complexity that characterizes their analysis. The development of simple yet effective control strategies – relying on few sensor readings and being robust to rough model descriptions – remains a challenge that needs to be tackled to bridge the gap with real-world experimental validations. This is especially true for what concerns the multi-link case (Giorgio et al. 2019). On a more theoretical level, formal proofs of stability for existing controllers are an open challenge – to be addressed using infinite dimensional control – when attacked in the proper continuum mechanics description of the flexible robot. Ultimately, dealing with the general case as expressed by (6) remains a largely unexplored topic.

Finally, one of the greatest challenges ahead for flexible robot community is establishing proper ways of connecting and integrating with soft robotics field. This needs to be done without falling into the perils of considering soft robots as a sub-class of flexible manipulators and vice-versa without looking at flexible manipulators as the old way of doing soft robotics. As already discussed above, articulated soft robots and flexible joint ones are particularly exposed to this fallacy, since they share both a similar mechanical structure and an identical mathematical formulation. As for now, almost no connection exists instead between continuum soft and flexible link community, despite the strong similarities between the two classes of robots. The development of accurate yet tractable 3D nonlinear models of structural deformations is a clear example of how the first community could inform the second (Grazioso et al. 2016), while control strategies developed for flexible link robots have the potential of informing effective controllers for continuum soft robots (Santina and Rus 2019). Of note, it is worth noticing that the problem of controlling continuum soft robots could come out to be even simpler that the one of controlling flexible link ones. Indeed, better collocation properties are expected, being the control action typically distributed along the structure (Santina et al. 2019).

Cross-References

- ▶ Aerial Robotic Manipulators
- ▶ Control of Aerial Robotic Manipulators
- ▶ Macro-Mini Manipulation
- ▶ Series Elastic Actuation
- ▶ Snake-Like and Continuum Robots
- ▶ Soft Robots
- ▶ Stiffness Control
- ▶ Variable Impedance Actuators

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